Week 3

1) Shortest Route for a Delivery Company

Application: Route optimization

Problem Statement: A delivery Company needs to deliver parcels to multiple cities and return to it's starting depot. Develop a python program using ACO to determine the shortest possible route that visits every city exactly once before returning to the depot

objective: Minimize the total travel distance.

Psudo code:

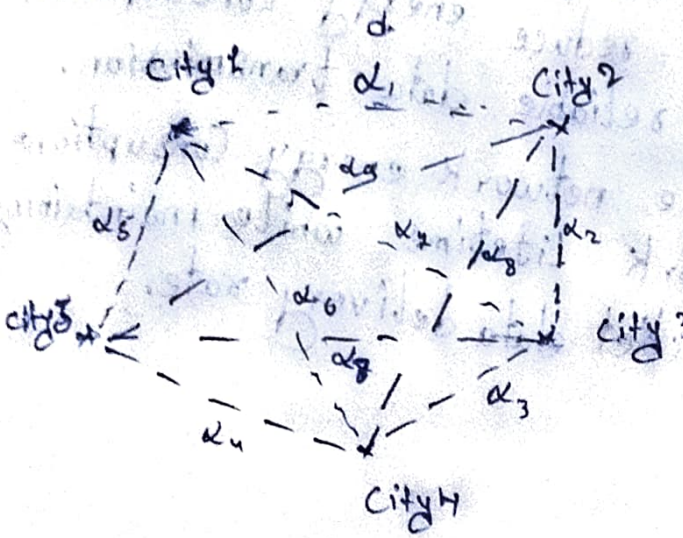
Begains:

~~Set number of ants~~

1. Initialize Parameter
   set number of ants, cities, pheromone values, evaporation ~~rat~~ rate, alpha, beta and iterations
2. Initialize Distance
   calculate distance between all cities.
   initialize pheromone values on all routes.
3. Construct Solutions.
   Place each ant at the starting depot.
   Select the next unvisited city based on pheromone strength and distance.
   Repeat until all cities are visited.
4. Complete the tour.
   Return each ant to the starting depot
   Calculate the total distance of each route
5. Update ~~pher~~ pheromones
   Evaporate old pheromones
   Deposit new pheromones based on route quality
   Keep the shortest route as the best Solution

6. Check Stopping Condition and output
   Repeate Step3-5 until maximum iterations are reached.
   Display the Shortest route and total distance

End.



30/10/26

output:

Best Route:
A -> B -> D -> E -> C -> A
Shortest Distance: 85

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30/10/26